CREDIT CARD FRAUD DETECTION PROJECT

1. Problem Statement

Title: Credit Card Fraud Detection using Machine Learning

Problem Statement:

Credit card fraud is a major challenge in the financial sector, leading to significant financial losses. The objective of this project is to develop a machine learning model that can accurately identify fraudulent transactions from a large dataset. The main challenge is that the dataset is highly imbalanced, with very few fraudulent transactions compared to legitimate ones. The goal is to build a model that can effectively classify transactions as legitimate or fraudulent while minimizing errors.

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2. Objectives

- To analyze credit card transaction data
- To handle class imbalance using undersampling
- To build a classification model using Logistic Regression
- To evaluate model performance using accuracy and other metrics

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3. Dataset Description

- Total transactions: ~284,807
- Features: 31

Columns:

- **Time** → Time of transaction
- **V1–V28**→Anonymized features
- **Amount** → Transaction amount
- **Class** → Target variable
- 0 → Legitimate
- 1 → Fraudulent

  Dataset is **highly imbalanced**

4. Solution Approach

Step 1: Import Libraries

- pandas, numpy
- sklearn (Logistic Regression, train_test_split, metrics)

Step 2: Load Dataset

- Read CSV file into DataFrame
- Check structure and class distribution

Step 3: Data Analysis

- Use `.describe()` and `.value_counts()`
- Identify imbalance in Class column

Step 4: Separate Classes

- ➢ Split into:
- Legit transactions
- Fraud transactions

Step 5: Handle Imbalance (Undersampling)

- ➢ Randomly select equal number of legit transactions
- ➢ Combine with fraud data
- ➢ Create balanced dataset

Step 6: Split Features & Target

- X → all columns except Class
- Y → Class column

Step 7: Train-Test Split

- 80% training, 20% testing
- Use stratify to maintain balance

Step 8: Feature Scaling

- Use StandardScaler
- Fit on training data, transform test data

Step 9: Model Training

- Use Logistic Regression
- Train using `fit()`

Step 10: Model Evaluation

- Predict using `predict()`
- Calculate accuracy for:
- Training data (~95%)
- Testing data (~91%)

5. Result

- Model successfully classified transactions
- Training Accuracy: ~95%
- Testing Accuracy: ~91%
- Shows good generalization with minimal overfitting

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6. Conclusion

The model was able to effectively detect fraudulent transactions after handling class imbalance using undersampling. Logistic Regression performed well with proper scaling and parameter tuning. However, in real-world scenarios, additional metrics like recall and precision should be considered to improve fraud detection performance.